

University of East London

School of Architecture, Computing and Engineering

Lab Report

CN6005 Artificial Intelligence

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Week 3: Comparing KNN & Logistic Regression Classifiers using Titanic Dataset

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- # Introduction:

This project predicts whether a Titanic passenger survived using four simple features: ticket class, sex, age and fare. Survival is a yes-or-no outcome, which makes it a good fit for binary classification. The first step is to check how many passengers survived and how many did not. The data is then prepared so the models can use it, by removing identifiers, converting sex to numbers and filling in missing ages. Next, a logistic regression model and a K-Nearest Neighbours (KNN) model are trained, and each one is tested on passengers it has not seen before. Finally, the two are compared to see which handles this dataset better and why.

- # Method:

The first step was choosing which information the models could actually use. Names, ticket numbers and passenger IDs are just labels, so they were left out, and sex was converted to binary form so the models could work with it. Some passengers had no age recorded, and rather than throw those rows away and lose data, the gaps were filled with the median age for logistic regression and the mean age for KNN. For KNN, the cabin and embarkation columns were also dropped, because most cabin values were missing and the port isn't a number. To check that the models weren't just memorising the answers, 30% of the data was held back for testing in both cases. For KNN, the number of neighbours was set to 3, so each passenger is classified by looking at the three most similar passengers in the training data. Both models were judged with a confusion matrix and a classification report, so it was possible to see not just how often they were right but what kind of mistakes they made.

- # Comparison:

Logistic regression fits one overall pattern across all passengers, while KNN classifies each passenger by looking at the three most similar passengers in the training data.

In terms of accuracy, logistic regression reached 77% and KNN reached about 70%, so logistic regression was better by about 7 percentage points. Since 62% of passengers died, predicting "died" for everyone would already score around

62%, so both models beat that baseline, logistic regression by about 15 points and KNN by about 8.

The confusion matrices show where those scores come from. Logistic regression made 74 correct predictions for deaths and 132 for survivors, with 62 wrong in total. KNN made 63 correct for deaths and 125 for survivors, with 80 wrong in total. Both models were better at identifying deaths than survivors, since deaths are the larger group. The survivor F1-score was 70% for logistic regression and 61% for KNN.

The dataset was left unbalanced, at roughly 62% died and 38% survived, and logistic regression still reached the higher accuracy and F1-score. This shows that balancing the dataset is not always the best option for improving accuracy and F1-score.